

pH & Kw 4

Essential Pre-Uni Chemistry J2.4

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Part A

(a)

Calculate the pH of a $0.012 \text{ mol dm}^{-3}$ solution of HCl.

Part B

(b)

Calculate the pH of a $0.030 \text{ mol dm}^{-3}$ solution of sulfuric acid. (For the purpose of this question, assume that sulfuric acid dissociates fully.)

Part C

(c)

Calculate the concentration of a solution of nitric acid with $\text{pH} = 2.1$.

Part D

(d)

Calculate the concentration of a solution of sulfuric acid with a pH of 4.7. (For the purpose of this question, assume that sulfuric acid dissociates fully.)

Question deck:

STEM SMART Chemistry Week 22

pH & Kw 5

Essential Pre-Uni Chemistry J2.5

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

Part A

(a)

Calculate the concentration of a solution of barium hydroxide with a pH of 9.5.

Part B

(b)

Calculate the pH of a $0.0800 \text{ mol dm}^{-3}$ solution of KOH.

Question deck:

STEM SMART Chemistry Week 22

pH & Kw 7

Essential Pre-Uni Chemistry J2.7

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

14 g of sulfuric acid is dissolved in 500 m³ of water. Calculate the pH of the resulting solution. (For the purpose of this question, assume that sulfuric acid dissociates fully.)

Question deck:

STEM SMART Chemistry Week 22

pH & Kw 8

Essential Pre-Uni Chemistry J2.8

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

100 cm³ of a solution of 0.750 mol dm⁻³ sulfuric acid is mixed with 400 cm³ of a solution of 0.300 mol dm⁻³ sodium hydroxide. Calculate the pH of the resulting mixture. Give your answer to 2 significant figures. (For the purpose of this question, assume that sulfuric acid dissociates fully.)

Question deck:

STEM SMART Chemistry Week 22

pH & Kw 9

Essential Pre-Uni Chemistry J2.9

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

50.0 cm³ of a solution of 0.200 mol dm⁻³ nitric acid is mixed with 200 cm³ of a solution of 0.160 mol dm⁻³ potassium hydroxide. Calculate the pH of the resulting mixture.

Question deck:

STEM SMART Chemistry Week 22

K_a & pK_a 2

Essential Pre-Uni Chemistry J3.2

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Species	$K_a / \text{mol dm}^{-3}$
Benzoic acid	6.3×10^{-5}
Hydrogen sulfide	8.9×10^{-8}
Iron(III)	6.0×10^{-3}
Methanoic acid	1.6×10^{-4}
Sulfuric(IV) acid	1.5×10^{-2}
Boric acid	5.8×10^{-10}

Calculate the concentration of a solution of benzoic acid with a pH of 3.2.

Question deck:

STEM SMART Chemistry Week 22

K_a & pK_a 4

Essential Pre-Uni Chemistry J3.4

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

The pK_a of ethanoic acid is 4.8. Calculate its K_a.

Question deck:

STEM SMART Chemistry Week 22

Ka & pKa 6

Essential Pre-Uni Chemistry J3.6

Subject & topics: Chemistry | Physical | Acids & Bases

Stage & difficulty: A Level P2

Calculate the K_a of an acid, HA, with a pH of 5.0 when its concentration is 0.20 mol dm^{-3} .

Question deck:

STEM SMART Chemistry Week 22

Ka & pKa 7

Essential Pre-Uni Chemistry J3.7

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

Species	$K_a / \text{mol dm}^{-3}$
Benzoic acid	6.3×10^{-5}
Hydrogen sulfide	8.9×10^{-8}
Iron(III)	6.0×10^{-3}
Methanoic acid	1.6×10^{-4}
Sulfuric(IV) acid	1.5×10^{-2}
Boric acid	5.8×10^{-10}

Part A

(a)

Calculate the pH of a 13.8 g dm^{-3} solution of methanoic acid.

Part B

(b)

240 cm^3 of hydrogen sulfide gas (at RTP) is dissolved in 500 cm^3 of water. Calculate the pH of the resulting solution.

Question deck:

STEM SMART Chemistry Week 22

Ka & pKa 9

Essential Pre-Uni Chemistry J3.9

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

Species	$K_a / \text{mol dm}^{-3}$
Benzoic acid	6.3×10^{-5}
Hydrogen sulfide	8.9×10^{-8}
Iron(III)	6.0×10^{-3}
Methanoic acid	1.6×10^{-4}
Sulfuric(IV) acid	1.5×10^{-2}
Boric acid	5.8×10^{-10}

Sulfur dioxide dissolves in water to give sulfuric(IV) acid: $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_2\text{SO}_3(\text{aq})$. Calculate the RTP volume of sulfur dioxide required to reduce the pH of a lake of volume 0.40 km^3 from 7.0 to 6.0.

Question deck:

STEM SMART Chemistry Week 22